Paradigm Justification Statement

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1. Research topic and context

Generative AI has entered Peruvian undergraduate coursework while local evidence is still emerging: a survey across four Peruvian universities associated ChatGPT use with students' ethical concerns and called for responsible-use programs (Acosta-Enriquez et al., 2024). In my courses I observe a recurring pattern of full delegation: students paste the assignment into a chatbot and submit the output without mediating it through their own judgment. Detectors and sanctions may curb this behavior, but they do not form the capacity to deliberate.

2. Preliminary research question

RQ1 (design): How can a conversational agent grounded in virtue ethics scaffold ethical deliberation when undergraduates at a Peruvian university delegate work to generative AI?

RQ2 (evaluation): Does forming autonomous judgment through this agent outperform restricting behavior, increasing metacognitive reflection and product integrity and reducing AI dependence relative to unrestricted and restrictive-normative conditions?

3. Philosophical position, research approach and justification

My philosophical position is **pragmatism**: knowledge is warranted by its usefulness in guiding action (Goldkuhl, 2012). Within it, **Design Science Research (DSR) is my research approach**, not my paradigm: knowledge of a problem and its solution is produced by building and evaluating an artifact, judged by its utility in context (Hevner et al., 2004). Because my question is prescriptive, my intended contribution is a nascent design theory: virtue-based design principles, instantiated in an agent and an annotated dataset, then empirically evaluated (Gregor & Hevner, 2013). Positivism cannot govern this study because it explains and predicts what already exists, whereas my object must first be designed; its hypothetico-deductive logic survives only inside the evaluation. Interpretivism would clarify how students experience delegation, useful design input, but it produces neither the artifact nor the comparative evidence an institution needs before adoption. Mixed methods (Fetters et al., 2013) is a methodology, not a position: it describes and explains rather than producing design principles, so it is my evaluation repertoire.

4. Implications

Data. A Spanish-language corpus of academic-integrity scenarios annotated for the four cardinal virtues plus honesty; agreement reported as Krippendorff's alpha per virtue, not aggregated (Hayes & Krippendorff, 2007). Outcomes: reflection coded with a four-category scheme (Kember et al., 2008); AI dependence with the DAI scale, developed with Peruvian university students (Morales-García et al., 2024); product integrity as a composite of log-based provenance and a blinded expert rubric, with content validity still to be established.

Methods. A FEDS Human Risk & Effectiveness strategy (Venable et al., 2016): formative, artificial episodes (expert validation, pilot annotation) precede a summative, naturalistic three-condition evaluation (sequential explanatory mixed methods). The intact section is the unit of assignment, with at least two sections per condition, multilevel models for clustering, a manipulation check for the restrictive condition, and cross-section contamination declared.

Positionality and ethics. As designer, instructor and evaluator of my own sections, I will obtain ethics approval, have consent collected by a third party and decoupled from grading, use blinded coders, and keep pseudonymised logs off public repositories.

Contribution and venues. Evaluated design principles and a Spanish-language resource for Latin America, for *Computers and Education: AI*, *Int. J. for Educational Integrity*, DESRIST and IEEE-RITA.

5. One doubt or tension

The more operational my definition of prudence, the higher inter-annotator agreement is likely to be, but the closer the construct drifts toward a rulebook, the very restrictive-normative logic I question. This trade-off is intrinsic to virtue ethics, because practical wisdom resists codification; it is not a design flaw. I will anchor annotation in expert-validated exemplar cases rather than rule-based definitions, reporting disagreement per virtue rather than simplifying the construct.

References

Acosta-Enriquez, B. G., Arbulú Ballesteros, M. A., Arbulu Perez Vargas, C. G., Orellana Ulloa, M. N., Gutiérrez Ulloa, C. R., Pizarro Romero, J. M., Gutiérrez Jaramillo, N. D., Cuenca Orellana, H. U., Ayala Anzoátegui, D. X., & López Roca, C. (2024). Knowledge, attitudes, and perceived ethics regarding the use of ChatGPT among generation Z university students. *International Journal for Educational Integrity, 20*, Article 10. https://doi.org/10.1007/s40979-024-00157-4

Fetters, M. D., Curry, L. A., & Creswell, J. W. (2013). Achieving integration in mixed methods designs—Principles and practices. *Health Services Research, 48*(6 Pt 2), 2134–2156. https://doi.org/10.1111/1475-6773.12117

Goldkuhl, G. (2012). Pragmatism vs interpretivism in qualitative information systems research. *European Journal of Information Systems, 21*(2), 135–146. https://doi.org/10.1057/ejis.2011.54

Gregor, S., & Hevner, A. R. (2013). Positioning and presenting design science research for maximum impact. *MIS Quarterly, 37*(2), 337–355. https://doi.org/10.25300/MISQ/2013/37.2.01

Hayes, A. F., & Krippendorff, K. (2007). Answering the call for a standard reliability measure for coding data. *Communication Methods and Measures, 1*(1), 77–89. https://doi.org/10.1080/19312450709336664

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly, 28*(1), 75–105. https://doi.org/10.2307/25148625

Kember, D., McKay, J., Sinclair, K., & Wong, F. K. Y. (2008). A four-category scheme for coding and assessing the level of reflection in written work. *Assessment & Evaluation in Higher Education, 33*(4), 369–379. https://doi.org/10.1080/02602930701293355

Morales-García, W. C., Sairitupa-Sanchez, L. Z., Morales-García, S. B., & Morales-García, M. (2024). Development and validation of a scale for dependence on artificial intelligence in university students. *Frontiers in Education, 9*, 1323898. https://doi.org/10.3389/feduc.2024.1323898

Venable, J., Pries-Heje, J., & Baskerville, R. (2016). FEDS: A framework for evaluation in design science research. *European Journal of Information Systems, 25*(1), 77–89. https://doi.org/10.1057/ejis.2014.36

AI-use disclosure: Claude (Anthropic) was used for language editing and for drafting the wording of this revision; the sources come from the author's prior research, and all methodological decisions are the author's.